PRODUCT LAYER

Project Overview

Project Overview Document

Fintech Social Intelligence & Financial Ecosystem Platform

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1. Introduction

This application is a regulated fintech-social ecosystem that combines:

- Verified financial advisory sentiment
- Community-driven engagement
- AI-powered financial intelligence
- Virtual investment simulation
- Real portfolio analytics
- Personal finance & UPI expense tracking

The core idea is to create a **single integrated platform where users can spend, save, invest, learn, and interact responsibly within a compliant financial environment**.

This is not just a trading app.
It is a **financial intelligence network powered by AI and regulated advisory participation**.

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2. Core Concept

The platform is built around four pillars:

1. **Trust & Regulation**
   Only SEBI-registered advisors can publish market sentiments.
   Compliance layers prevent manipulation or misleading claims.

2. **Intelligent Finance Management**
   Users can connect portfolios and UPI accounts to get AI-driven insights on:

    - Investments
    - Spending
    - Savings behavior
    - Risk exposure

3. **Learning & Simulation**
   A virtual investment lab allows users to practice investing with virtual money.

4. **Social Financial Community**
   A structured financial social network where:
    - Advisors share regulated insights
    - Users interact responsibly
    - Non-market discussions happen in a separate safe zone

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3. What This Application Will Be

This application will function as:

1. A Regulated Market Sentiment Platform

- SEBI-verified advisors publish market outlooks.
- AI aggregates sentiment across sectors and indices.
- Users can engage but cannot manipulate sentiment metrics.

2. A Financial Social Network

- Users share financial journeys, savings tips, lifestyle discussions.
- Groups and forums for personal finance topics.
- Gamified reputation system.

3. A Virtual Investment Lab

- Users receive virtual capital.
- Practice trading across asset classes.
- Participate in leaderboards and competitions.
- Backtest strategies.

4. A Portfolio Intelligence Engine

- Users connect real brokerage accounts.
- AI analyzes:
    - Diversification
    - Risk concentration
    - Rebalancing needs
    - Tax optimization opportunities

5. A Personal Finance Manager

- Users connect UPI-linked bank accounts.
- Automatic expense categorization.
- Budget tracking.
- Subscription detection.
- Financial health scoring.

6. An AI Financial Assistant

- Chatbot answers finance queries.
- Behavioral finance detection.
- Cash flow forecasting.
- Emergency fund analysis.

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4. Target Users

User Type	Primary Use Case
Beginner Investors	Learn via virtual lab & AI assistant
Retail Investors	Track portfolio & advisor sentiment
SEBI Advisors	Publish insights & monetize knowledge
Salaried Professionals	Track expenses & optimize savings
Financial Enthusiasts	Participate in community discussions

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5. Primary Use Cases

Use Case 1: Retail Investor Tracking Portfolio

- Connects brokerage account.
- AI detects 65% exposure to one sector.
- Suggests diversification.
- Compares holdings with advisor sentiment.
- Provides rebalancing suggestions.

Outcome: Smarter, data-backed investing decisions.

Use Case 2: Beginner Learning Investment

- Starts with ₹10L virtual capital.
- Trades simulated stocks.

- Views risk analytics.
- Learns via advisor courses.
- Uses AI chatbot for clarification.

Outcome: Risk-free practical learning.

Use Case 3: Salaried Professional Managing Expenses

- Connects UPI bank account.
- AI categorizes expenses.
- Detects overspending in dining.
- Suggests savings adjustment.
- Forecasts month-end balance.

Outcome: Improved financial discipline.

Use Case 4: SEBI Advisor Monetizing Knowledge

- Verified via SEBI registration.
- Posts market insights.
- Launches paid subscription.
- Uploads video course.
- Engages audience via premium room.

Outcome: Digital advisory monetization channel.

Use Case 5: Financial Health Optimization

- AI calculates financial health score.
- Combines:
    - Spending pattern
    - Investment allocation
    - Savings ratio
- Suggests:
    - SIP increase
    - Expense reduction
    - Emergency fund target

Outcome: Holistic financial improvement.

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6. Key Differentiators

1. Advisor-Only Market Posting
   Prevents misinformation and pump-and-dump behavior.

2. AI-Driven Cross-Analysis
   Portfolio + Sentiment + Expenses integrated together.

3. Separation of Market & Social Content
   Reduces manipulation risk.

4. Virtual Lab Integration
   Education + Practice in one system.

5. Personal Finance + Investment Combined
   Covers full financial lifecycle.

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7. Main Concept Architecture (Simplified)

User Data Streams:

- Portfolio Data
- Expense Data
- Market Sentiment Data
- Social Engagement Data

AI Layer Processes:

- Risk analytics
- Behavior analysis
- Cash flow forecasting
- Sentiment aggregation

Output:

- Financial health score
- Rebalancing suggestions
- Budget alerts
- Learning recommendations

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8. Business Vision

To build India's first:

Regulated AI-powered financial social intelligence platform
that merges:

- Advisory
- Community

- Simulation
- Portfolio analytics
- Expense intelligence

Into one scalable fintech ecosystem.

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9. Long-Term Vision

- Become a trusted advisory sentiment index.
- Offer institutional integrations.
- Build robo-advisory layer.
- Expand into financial planning services.
- Introduce AI-based goal planning.

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10. Conclusion

This platform is designed to:

- Improve financial literacy
- Encourage responsible investing
- Prevent misinformation
- Enable advisor monetization
- Provide AI-driven personal financial intelligence

It transforms financial decision-making from reactive to intelligent, structured, and data-driven.

Internal Feature Mapping Table

COMPLETE INTERNAL FEATURE MAPPING TABLE

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1. User Management & Identity Layer

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
User Registration	Creates user profile	None	Portfolio, Social, Payments, AI	Risk profiling model	User signs up via mobile	KYC required before portfolio linking
Role Management	Assigns roles (Normal / SEBI Advisor / Admin)	Registration	Content permissions, Monetization	None	Advisor gets verified badge	Advisor verification mandatory
KYC Verification	Identity verification	User Registration	Portfolio linking, Payments	Fraud detection model	PAN + Aadhaar verification	SEBI & AML compliance
Risk Profiling	Captures investment risk appetite	KYC	Portfolio AI, Recommendations	Risk scoring ML model	User marked as Moderate Risk	Must show disclaimer
Device Authentication	Secure login	Registration	All features	Behavioral anomaly detection	New device login flagged	Security logging required

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2. Portfolio Connection & AI Insight Engine

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
Broker API Integration	Fetches user holdings	KYC	Portfolio AI	Data validation model	Zerodha portfolio synced	Must use secure OAuth
Manual Portfolio Upload	CSV upload	Registration	Portfolio AI	Data cleaning model	User uploads MF portfolio	Data encryption required
Portfolio Snapshot Engine	Daily valuation calculation	Broker Sync	Analytics	Trend detection AI	Daily NAV updated	Historical audit logs
Asset Allocation Analyzer	Sector & asset breakdown	Snapshot Engine	Risk Alerts	Allocation ML	45% in Banking flagged	Informational only
Risk Exposure Detector	Identifies overexposure	Allocation Analyzer	Notifications	Risk ML model	High small-cap exposure	No direct buy/sell advice
Rebalancing Suggestions	Suggest diversification	Risk Detector	Notifications	Portfolio AI Agent	Suggests 60/40 equity/debt	Must show advisory disclaimer
Portfolio Health Score	0–100 score	Risk + Volatility data	Dashboard	Scoring model	Health Score: 72	Non-guaranteed projection

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3. Fintech Social Community (Market Sentiment)

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
Advisor Market Post	SEBI advisors post market sentiment	Role Management	Sentiment Feed	Sentiment NLP model	“Banking sector bullish”	Advisor ID displayed
Sentiment Tagging	Bullish/Bearish tagging	Advisor Post	Analytics	NLP classifier	Classified as Bullish	No price targets allowed
Market Feed Timeline	Shows advisor posts	Advisor Post	Engagement	Ranking algorithm	Trending posts	Anti-manipulation monitoring
Engagement Engine	Likes/comments (controlled)	Market Feed	Reputation Score	Bot detection AI	200 likes on post	Prevent artificial inflation
Manipulation Detection	Detect pump/dump	All posts	Admin Alerts	Anomaly detection	Sudden 1000 bullish posts	Mandatory flagging
Advisor Accuracy Tracker	Track past sentiment vs outcome	Market Data	Reputation System	Performance ML	65% accuracy rate	Historical transparency

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4. General Social Community (Non-Market)

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
Public Posts	General financial discussions	User Role	Social Graph	Content moderation AI	“How to save tax?”	No stock tips allowed
Hashtag Engine	Topic grouping	Posts	Discover Feed	Semantic clustering	#FinancialPlanning	Content filtering
User Follow System	Follow users	Registration	Feed Personalization	Recommendation AI	Follow advisor	Data privacy compliance
Direct Messaging	User-to-user chat	Follow System	Notifications	Spam detection AI	DM financial query	Encrypted messages
Community Moderation	Flag inappropriate content	All posts	Admin Panel	Toxicity detection	Remove misleading claim	Must log removals

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5. Virtual Investment Lab (Paper Trading)

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
Virtual Wallet	Assign demo capital	Registration	Trade Engine	None	₹10L demo capital	Not real money
Trade Simulation Engine	Execute paper trades	Market Data	Performance Tracking	Strategy AI	Buy 100 Reliance demo	Clear “Simulation” label
Strategy Builder	Create investment strategies	Trade Engine	AI Insights	Strategy ML	Momentum strategy	Educational disclaimer
Leaderboard	Rank users	Performance Data	Gamification	Ranking algorithm	Top 10 traders	No financial reward
AI Performance Review	Analyze trade mistakes	Trade Data	User Dashboard	Trade pattern ML	High risk behavior flagged	No real advisory

6. AI Financial Chatbot System

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
Financial Q&A Bot	Answer finance queries	Knowledge Base	User Dashboard	LLM + RAG	“What is SIP?”	No direct stock recommendation
Portfolio AI Assistant	Analyze user holdings	Portfolio Engine	Notifications	Portfolio AI Agent	Suggest rebalancing	Must show disclaimer
Expense AI Coach	Budget suggestions	Expense Tracking	Alerts	Spending ML	“Reduce dining expense”	Informational only
Sentiment AI Analyzer	Analyze market posts	Market Feed	Analytics	NLP model	Bullish score 0.78	Anti-manipulation

7. Advisor Monetization Engine

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
Course Marketplace	Advisors sell courses	Role Management	Payments	Recommendation AI	Options Trading Course	SEBI compliance required
Subscription Engine	Paid access to advisor feed	Payment Service	Access Control	Churn prediction ML	₹999/month plan	Auto-renew transparency
Revenue Split Engine	Commission calculation	Payment	Wallet System	Fraud detection AI	80/20 split	Tax compliance
Advisor Analytics Dashboard	Performance metrics	Data Warehouse	Reputation Score	Performance AI	2,000 subscribers	Transparent reporting

8. Personal Finance & UPI Tracking

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
UPI Transaction Sync	Fetch bank expenses	KYC	Expense Dashboard	Categorization AI	Swiggy → Food	RBI compliance
Auto Expense Categorization	Classify spending	UPI Sync	Budget Engine	NLP + ML	Uber → Transport	Data encryption
Budget Planning Tool	Monthly budget allocation	Expense Data	Alerts	Forecasting ML	₹10K food limit	Informational only
Savings Goal Tracker	Track savings	Expense Engine	Notifications	Savings ML	Goal ₹2L	Privacy compliance

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9. Analytics & Reputation System

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
Advisor Reputation Score	Score advisors	Sentiment + Accuracy	Feed Ranking	Performance ML	78/100 rating	Must disclose formula
User Engagement Score	Rank active users	Social Activity	Leaderboard	Behavior ML	Top contributor	Anti-gaming system
Trend Detection Engine	Detect trending stocks	Market Posts	Alerts	Time-series ML	IT sector trending	Manipulation guard

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10. Admin & Compliance Control

Feature	What It Does	Depends On	Feeds Into	AI Involvement	Example	Compliance Notes
Content Flagging	Review flagged posts	Moderation AI	Admin Panel	NLP detection	Remove false claim	Mandatory logging
Audit Log System	Track all actions	All modules	Regulatory Reports	Anomaly detection	Track edit history	SEBI-ready export
Manipulation Watchlist	Detect suspicious activity	Sentiment Engine	Admin Alerts	Pattern ML	Sudden bullish spike	Required safeguard
Compliance Report Generator	Generate reports	Data Warehouse	External Reporting	Aggregation engine	Monthly advisor report	Regulator export

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Cross-Feature Interaction Summary

Core Feature	Directly Impacts
Portfolio Engine	AI Bot, Analytics, Notifications
Sentiment Engine	Reputation Score, Compliance, Analytics
Payment System	Advisor Monetization, Subscription Access
Expense Tracking	AI Coach, Savings Goals
AI Agent Layer	Portfolio, Expense, Sentiment, Moderation
Data Warehouse	Dashboards, AI Training, Reports
Risk Profiling	Portfolio AI, Chatbot Personalization

Example Full System Interaction Scenario

User connects portfolio →
Portfolio Engine calculates allocation →
Risk Detector flags overexposure →
AI Portfolio Agent generates insight →
Notification sent →
User asks chatbot for explanation →
AI references portfolio data →
User subscribes to advisor course →
Payment Engine processes →
Revenue Split recorded →
Advisor analytics updated →
Data Warehouse logs event →
Reputation score recalculated

End-to-End Feature Flow Document

1. USER AUTHENTICATION FLOW

1.1 User Registration Flow

Actor: New User / Advisor

Step 1: User submits registration form
Inputs:

- Name
- Email
- Phone
- Password
- Role (User / Advisor)

Step 2: Backend Validation

- Email uniqueness
- Phone uniqueness
- Password strength
- Role validity

Step 3: If Advisor selected

- Require SEBI Registration Number
- Store as “Pending Verification”

Step 4: Create User Record
Tables affected:

- users
- advisor_profiles (if advisor)

Step 5: Generate JWT Token

Step 6: Return success response

Edge Cases:

- Duplicate email
- Invalid SEBI number format
- Weak password

Failure Handling:
Return validation error JSON.

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2. ADVISOR VERIFICATION FLOW

Actor: Admin

Step 1: Advisor submits verification documents
Step 2: Admin reviews
Step 3: Admin approves/rejects

If Approved:

- advisor_profiles.verification_status = approved
- Enable Market Posting capability

If Rejected:

- Store rejection reason
- Notify advisor

Notification triggered:

- “Your verification status has changed”

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3. MARKET SENTIMENT POST FLOW (CORE COMPLIANCE FLOW)

Actor: Verified Advisor

Step 1: Advisor creates market post
Fields:

- Title
- Content
- Asset Type
- Sentiment (Bullish/Bearish/Neutral)
- Risk Level
- Disclaimer

Step 2: Backend Validation

- Advisor verified
- Mandatory disclaimer present
- No prohibited language

Step 3: AI Compliance Agent Triggered
Checks:

- Market manipulation language
- Unrealistic claims
- Guaranteed returns wording
- Price target without context

If AI flags high risk:

- Mark post as “Under Review”
- Send to admin moderation queue

If Safe:

- Save in market_posts
- Publish to feed

Step 4: Followers Notification

- Push notification
- In-app notification

Database Impact:

- market_posts
- notifications
- audit_logs

Edge Cases:

- Advisor suspended
- Duplicate post
- AI service unavailable (fallback: manual moderation)

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4. MARKET COMMENT FLOW

Actor: Any User

Step 1: User comments on post
Step 2: Validate no direct investment advice
Step 3: AI quick toxicity scan
Step 4: Save to market_comments
Step 5: Notify advisor

Edge Cases:

- Spam detection
- Excessive commenting rate

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5. COMMUNITY POST FLOW

Actor: Any User

Step 1: Create post
Step 2: Validate no direct stock recommendation
Step 3: AI content moderation
Step 4: Store in community_posts
Step 5: Distribute in feed algorithm

Community Feed Ranking Factors:

- Engagement score
- Recency
- Relevance
- User connections

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6. PORTFOLIO CONNECT FLOW (REAL PORTFOLIO)

Actor: User

Step 1: User selects broker
Step 2: OAuth authentication (external broker)
Step 3: Receive API token
Step 4: Encrypt and store token
Step 5: Fetch holdings
Step 6: Store in portfolios + portfolio_assets

Step 7: Trigger AI Portfolio Analysis

If Broker API fails:

- Retry 3 times
- Notify user

Security:

- Tokens encrypted
- Never stored in plain text

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7. MANUAL PORTFOLIO ENTRY FLOW

Actor: User

Step 1: Add asset manually
Step 2: Enter:

- Asset name
- Quantity
- Buy price

Step 3: Store in portfolio_assets
Step 4: Trigger AI Analysis

---

8. AI PORTFOLIO ANALYSIS FLOW

Trigger: Portfolio updated

Step 1: Fetch portfolio data
Step 2: Calculate:

- Total value
- Asset allocation
- Sector allocation
- Volatility index
- Beta exposure

Step 3: Risk Scoring Algorithm
Inputs:

- Equity %
- Small cap exposure
- Crypto exposure
- Concentration ratio

Step 4: Diversification Score

Step 5: Generate Rebalance Suggestions

Step 6: Store in portfolio_ai_analysis

Step 7: Notify user if high risk

Edge Case:

- AI service timeout → fallback rule-based scoring

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9. VIRTUAL INVESTMENT LAB FLOW

Actor: User

Step 1: User creates virtual account

- Initial virtual balance assigned

Step 2: Place trade

- Select asset
- Enter quantity
- Buy/Sell

Step 3: Fetch live price API
Step 4: Validate sufficient virtual balance
Step 5: Execute trade simulation
Step 6: Update virtual_trades
Step 7: Update balance

Step 8: Update leaderboard

Edge Case:

- Price API unavailable → block trade

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10. LEADERBOARD FLOW

Trigger:

- Trade completed
- Portfolio value change

System:

- Recalculate total return
- Rank users
- Update leaderboard cache

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11. AI CHATBOT FLOW

Actor: User

Step 1: User asks question
Step 2: Determine query type:

- Portfolio related
- General finance
- Expense
- Market

Step 3: Context Enrichment

- Pull user portfolio (if required)
- Pull expense summary (if required)

Step 4: Send structured prompt to AI
Step 5: Receive response
Step 6: Save chat log
Step 7: Return response

Rate Limiting:

- Free users limited daily

Edge Case:

- Inappropriate question → filtered

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12. UPI EXPENSE TRACKING FLOW

Actor: User

Step 1: Connect UPI
Step 2: OAuth / API authentication
Step 3: Fetch transactions

Step 4: AI Categorization Engine
Categories:

- Food
- Travel
- Investment
- EMI
- Utilities

Step 5: Store in transactions table

Step 6: Budget Engine

- Compare monthly spend vs limit

Step 7: Trigger alerts if exceeded

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13. COURSE PURCHASE FLOW

Actor: User

Step 1: Select course
Step 2: Redirect to payment gateway
Step 3: Payment success webhook
Step 4: Create enrollment record
Step 5: Unlock content
Step 6: Notify advisor

Failure:

- Payment failed → show retry option

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14. ADVISOR SUBSCRIPTION FLOW

Actor: User

Step 1: Subscribe to advisor
Step 2: Payment gateway recurring plan
Step 3: Create subscription record
Step 4: Grant premium access

Cron Job:

- Check subscription expiry
- Revoke access if expired

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15. NOTIFICATION FLOW

Triggers:

- New market post
- Comment
- Subscription
- Budget alert
- High risk portfolio

Process:

- Create notification record
- Push via Firebase
- Update badge count

---

16. REPORT & MODERATION FLOW

Actor: User

Step 1: Report post
Step 2: Save report
Step 3: Admin review queue

Admin Actions:

- Remove post
- Suspend user
- Warn user

Audit:

- All actions logged

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17. ADMIN PANEL FLOW

Admin Capabilities:

- Approve advisors
- Moderate posts
- View flagged content
- View audit logs
- Suspend users
- View analytics

All actions logged in audit_logs.

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18. DATA SYNC & BACKGROUND JOB FLOW

Scheduled Jobs:

- Portfolio sync
- Subscription check
- Expense monthly summary
- AI reanalysis
- Leaderboard refresh

Queue System:
Redis-based queues
Retry mechanism for failures

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19. FAILURE MANAGEMENT FLOW

If AI fails:

- Fallback rules

If Broker fails:

- Retry + notify

If Payment webhook fails:

- Retry webhook

If DB error:

- Log + alert DevOps

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20. SECURITY FLOW

On every request:

- JWT validation
- Role check
- Rate limit check
- Input sanitization
- XSS filter
- SQL injection prevention

Sensitive data:

- Encrypted at rest
- Encrypted in transit

COMPLETE FLOW SUMMARY

User Layer
↓
Flutter App
↓
API Gateway (Laravel)
↓
Business Logic Layer
↓
AI Layer + Compliance Engine
↓
Database Layer
↓
Notification Engine

FRONTEND (FLUTTER)

Wireframes

1. WIREFRAMES (STRUCTURAL BLUEPRINTS)

These are logical wireframes (component-level breakdown).

1.1 AUTHENTICATION MODULE

A. Login Screen

Components:

- App Logo
- Email / Mobile Input
- Password Input
- Login Button
- Forgot Password
- Register CTA

State Cases:

- Loading spinner
- Error state
- OTP verification modal (if required)

B. KYC Upload Screen

Components:

- PAN Input
- Aadhaar Input
- Document Upload (Front / Back)
- Selfie Upload
- Submit Button
- Status Indicator (Pending / Verified / Rejected)

1.2 HOME DASHBOARD (Role-Based)

Bottom Navigation:

1. Home
2. Portfolio
3. Community
4. Lab
5. Profile

A. Home (Normal User)

Sections:

- Portfolio Health Card
- AI Insight Summary Card
- Expense Snapshot
- Trending Market Sentiment
- Suggested Advisors

- Quick Actions (Add Portfolio / Ask AI / Set Budget)

B. Home (Advisor)

Sections:

- Post Market Sentiment Button
- Subscriber Count
- Revenue Summary
- Engagement Analytics
- Course Performance
- Compliance Alerts

1.3 MARKET SENTIMENT FEED

Components:

- Sentiment Filter (Bullish / Bearish / All)
- Stock Search
- Feed List (Card-based)

Post Card:

- Advisor Name + Verified Badge
- Sentiment Tag
- Text Content
- Accuracy % (Historical)

- Like / Comment
- Report Button

Comment Drawer:

- Threaded Comments
- Reply
- AI Summary (optional)

1.4 GENERAL COMMUNITY FEED

Components:

- Post Composer
- Media Upload
- Hashtag Tagging
- Like / Comment / Share

Strict filter:

- Market keywords trigger warning

1.5 PORTFOLIO DASHBOARD

Sections:

- Total Portfolio Value
- Day Change
- Asset Allocation Pie Chart

- Sector Exposure Chart
- Risk Score Indicator
- AI Recommendation Cards
- Connect Broker Button

Asset Detail Page:

- Holding Details
- Performance Chart
- AI Insight

1.6 EXPENSE TRACKING DASHBOARD

Sections:

- Monthly Spend Total
- Category Breakdown Chart
- Budget Utilization Bar
- AI Expense Advice Card
- Connect UPI Button
- Savings Goal Progress

1.7 VIRTUAL LAB

Sections:

- Virtual Balance

- Active Trades
- Market Watchlist
- Place Trade Button
- Leaderboard
- AI Trade Review

Trade Screen:

- Stock Search
- Quantity
- Buy/Sell
- Stop Loss
- Confirm

1.8 AI CHATBOT SCREEN

Components:

- Chat History
- Context Indicator (Portfolio-aware / Expense-aware)
- Suggestion Chips
- Send Button
- Disclaimer Banner

Supports:

- Portfolio queries
- Budget advice

- Market explanation
- Education content

1.9 ADVISOR MONETIZATION PANEL

Sections:

- Create Course
- Upload Content
- Subscription Plans
- Earnings Overview
- Payout Request Button

SCREEN FLOW (NAVIGATION ARCHITECTURE)

2. SCREEN FLOW (NAVIGATION ARCHITECTURE)

Below is high-level navigation logic.

2.1 ONBOARDING FLOW

Splash → Login/Register → KYC → Risk Profiling → Home

If Advisor:
→ Verification Upload → Pending Status → Approval → Advisor Dashboard

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2.2 PORTFOLIO FLOW

Home
→ Portfolio
→ Connect Broker
→ OAuth
→ Sync
→ Dashboard
→ AI Analysis

2.3 MARKET POST FLOW (Advisor)

Home
→ Create Sentiment
→ Write Post
→ AI Compliance Check
→ Publish
→ Feed Update
→ Engagement

2.4 UPI EXPENSE FLOW

Profile
→ Connect UPI
→ Bank Authorization
→ Transaction Sync
→ Auto Categorization
→ Dashboard Update
→ AI Advice

2.5 VIRTUAL LAB FLOW

Lab
→ Add Stock
→ Place Trade
→ Confirm
→ Update Virtual Wallet
→ Leaderboard

2.6 COURSE PURCHASE FLOW

Advisor Profile
→ View Course
→ Subscribe
→ Payment
→ Confirmation
→ Unlock Content

SEQUENCE DIAGRAMS (LOGICAL INTERACTION FLOWS)

3. SEQUENCE DIAGRAMS (LOGICAL INTERACTION FLOWS)

Textual technical flow for Flutter implementation.

3.1 Portfolio AI Insight Flow

User → Portfolio Screen
→ App calls Portfolio API
→ Backend returns holdings
→ App displays allocation
→ App triggers AI Insight API
→ Backend AI returns recommendations
→ Display Insight Card

Error case:
If AI fails → show fallback message

3.2 Sentiment Post Publishing

Advisor → Create Post
→ Flutter validates
→ Send to Backend
→ Backend sends to Compliance AI
→ AI returns approval
→ Store in DB
→ Push Notification Service
→ Feed refresh

If AI rejects:
→ Show reason
→ Allow edit

3.3 UPI Expense Categorization

User → Connect UPI
→ Backend sync transactions
→ Backend sends to AI Categorizer
→ Categorized response
→ Store categorized transactions
→ Flutter fetch dashboard

3.4 Chatbot Portfolio Context Flow

User opens Chat
→ App sends context flag (portfolio=true)
→ Backend injects portfolio data
→ AI generates response
→ Return structured JSON
→ Flutter renders response card

STATE MANAGEMENT PLAN (FLUTTER ARCHITECTURE)

4.1 GLOBAL STATE MODULES

State Module	Responsible For
AuthProvider	Login, token
UserProvider	Profile data
PortfolioProvider	Holdings & health score
MarketFeedProvider	Sentiment feed
CommunityProvider	General posts
LabProvider	Virtual trades
ExpenseProvider	UPI transactions
ChatProvider	AI chat state
PaymentProvider	Subscription state

4.2 STATE LIFECYCLE RULES

Authentication:

- Persist JWT securely
- Refresh token auto

Portfolio:

- Cached locally
- Auto refresh on pull-to-refresh

Market Feed:

- Paginated

- Optimistic like updates

Chat:

- Maintain conversation ID
- Context memory per session

Expense:

- Sync manually + auto nightly refresh

4.3 ERROR HANDLING STRATEGY

Every API response must handle:

- Loading state
- Empty state
- Error state
- Retry option

4.4 PERFORMANCE STRATEGY

- Lazy loading lists
- Image caching
- Chart virtualization
- Pagination for feeds
- WebSocket for real-time sentiment updates

4.5 SECURITY STRATEGY (FRONTEND)

- Store tokens in secure storage
- Obfuscate build
- SSL pinning
- Prevent screenshot on sensitive screens
- Biometric lock option

4.6 OFFLINE STRATEGY

Cache:

- Portfolio last snapshot
- Feed (last 20 posts)
- Expense summary

If offline:

- Disable trade simulation
- Allow chat draft save

FINAL FRONTEND MODULE STRUCTURE

```
lib/
 modules/
 auth/
 home/
 portfolio/
 market_feed/
 community/
 lab/
 expenses/
 chat/
 advisor/
 core/
 networking/
 state/
 models/
 utils/
 theme/
```

BACKEND

Detailed System Architecture

1. Architecture Overview

Layer	Technology	Purpose	Notes
Frontend	Flutter (iOS + Android)	Mobile application UI	Cross-platform development
API Gateway	Laravel (PHP)	Central request handling	Authentication, routing, rate limiting
Microservices	Laravel + Python	Modular business logic	Independent scaling
Database	PostgreSQL / MySQL	Structured data storage	Financial & social segregation
Cache	Redis	Performance optimization	Leaderboards, summaries
Queue	Redis / SQS	Async processing	AI tasks, notifications
AI Layer	Python + LLM APIs	Financial intelligence engine	NLP, forecasting, analytics
Cloud	AWS / GCP	Infrastructure hosting	Auto-scaling & monitoring

---

2. High-Level Layered Architecture

Layer	Components	Responsibility	Connected To
Presentation Layer	Flutter UI	User interaction	API Gateway
API Layer	Laravel Controllers	Route requests	Microservices
Business Layer	Service Classes	Core logic	Database + AI
AI Layer	Python AI Service	Analysis & predictions	Queue + DB
Data Layer	SQL Database	Persistent storage	All Services
Integration Layer	Broker APIs, Bank APIs, Payment Gateway	External data	Portfolio & UPI Services

3. Frontend Architecture (Flutter)

Module	Purpose	State Management	Real-Time Support
Authentication	Login, KYC	Riverpod/Bloc	No
Market Sentiment Feed	Advisor posts	Riverpod	WebSockets
Social Community	User posts & comments	Riverpod	WebSockets
Portfolio Dashboard	Holdings & analytics	Bloc	Background refresh
Virtual Lab	Simulated trading	Bloc	Real-time market feed
AI Insights	Risk & recommendations	Riverpod	Async refresh
Expense Dashboard	Budget & spending	Bloc	Background sync
Advisor Dashboard	Course & revenue management	Riverpod	No

4. Backend Microservices Architecture

Microservice	Core Responsibilities	Database Tables	External Integrations
Auth Service	Login, roles, KYC	users, roles, kyc_documents	KYC Provider
Advisor Service	Sentiment posts, disclosures	advisors, sentiments	SEBI validation
Social Service	Posts, groups, reputation	social_posts, comments, groups	CDN
Portfolio Service	Broker sync, holdings	portfolios, trades_real	Zerodha, Groww, Upstox

Virtual Lab Service	Simulated trades	trades_virtual, leaderboard	Market Data API
Finance Service	UPI sync, budgeting	transactions, budgets	RBI Account Aggregator
AI Service	Risk analysis, NLP, forecasts	ai_logs	LLM APIs
Compliance Service	Content monitoring	compliance_logs	NLP Engine
Payment Service	Subscriptions & courses	payments, subscriptions	Razorpay / Stripe
Notification Service	Alerts & emails	notification_logs	Firebase

---

5. AI Architecture

AI Component	Function	Input	Output	Trigger Mechanism
NLP Moderation Engine	Detect manipulation	Advisor Posts	Flag / Approval	Post submission
Sentiment Aggregator	Calculate bullish/bearish ratio	Advisor Sentiments	Sentiment Index	Post publish
Portfolio Risk Model	Diversification scoring	Holdings	Risk Score	Portfolio sync
Expense Categorization Model	Merchant tagging	Transaction description	Category	UPI sync
Cash Flow Forecast Model	Predict balance	Expense trends	Forecast graph	Daily batch job
Behavioral Finance Agent	Detect impulsive trades	Trade logs	Risk alert	Trade event
Tax Optimization Agent	Suggest deductions	Portfolio + Expense	Tax insights	Monthly batch
AI Chatbot	Financial query response	User question	AI answer	Real-time request

6. Data Flow Architecture

6.1 Portfolio Sync Flow

Step	Action	Service Involved	Output
1	User OAuth login	Portfolio Service	Access Token
2	Fetch holdings	Broker API	Raw portfolio data
3	Store encrypted data	Database	Portfolio record
4	Send to AI engine	Queue	Risk computation
5	Store analytics	Database	Risk metrics
6	Notify user	Notification Service	Alert

6.2 UPI Expense Tracking Flow

Step	Action	Service	Output
1	User grants consent	Finance Service	AA token
2	Fetch transactions	Bank API	Raw transactions
3	AI categorization	AI Service	Tagged expenses
4	Update budgets	Finance Service	Budget recalculation
5	Trigger alerts	Notification Service	Overspending alert

6.3 Advisor Sentiment Flow

Step	Action	Service	Output
1	Advisor creates post	Advisor Service	Draft post
2	NLP scan	AI Service	Flag/Approved

3	Compliance log	Compliance Service	Audit record
4	Publish	Social Service	Live post
5	Aggregate sentiment	AI Service	Sentiment index update

7. Database Architecture

Data Category	Tables	Encryption Level	Access Restriction
User Data	users, roles	Standard encryption	Role-based
Financial Portfolio	portfolios, trades_real	AES-256	Strict isolation
Virtual Lab	trades_virtual	Standard	User only
Expense Data	transactions, budgets	AES-256	User only
Sentiment Data	sentiments	Standard	Public read
AI Logs	ai_logs	Encrypted	Admin only
Compliance Logs	compliance_logs	Encrypted	Admin only

8. Security Architecture

Security Layer	Implementation	Purpose
Authentication	JWT / Sanctum	Secure login
Authorization	Role-based middleware	Controlled access
Data Encryption	AES-256	Protect financial data
OAuth2	Broker & bank integration	Secure data sync
Rate Limiting	API throttling	Prevent abuse
Audit Trails	Compliance logs	Regulatory safety
Consent Management	AA Framework	User data control

9. Infrastructure Architecture (AWS Example)

Component	AWS Service	Purpose
Application Hosting	EC2 / ECS	Backend hosting
Database	RDS	Managed SQL DB
Cache	Elasticache (Redis)	Speed optimization
Storage	S3	File & course storage
CDN	CloudFront	Fast content delivery
Queue	SQS	Async processing
Monitoring	CloudWatch	System monitoring
Auto Scaling	ASG	Handle load spikes

---

10. Scalability Strategy

Component	Strategy
API Layer	Stateless design
AI Processing	Queue-based async execution
Database	Indexing & partitioning
Media	CDN caching
Microservices	Independent horizontal scaling

---

11. Failover & Backup Strategy

Risk	Mitigation Strategy
Database failure	Multi-AZ RDS

Traffic spike	Auto-scaling
AI service crash	Separate container restart
Data corruption	Daily backups
Security breach	Audit logging & anomaly detection

12. System Interconnectivity Matrix

Module	Portfolio	UPI	AI	Sentiment	Social	Lab
Portfolio Service	—	Indirect	Direct	Compare exposure	No	Strategy test
Finance Service	No	—	Direct	No	No	Budget simulation
AI Service	Direct	Direct	—	Direct	Indirect	Direct
Advisor Service	Indirect	No	Direct	—	Direct	Backtest
Virtual Lab	Indirect	No	Direct	Indirect	Leaderboard	—

MICROSERVICE ARCHITECTURE DESIGN

1. FULL MICROSERVICE ARCHITECTURE DESIGN

This is designed for:
• Large-scale fintech platform
• Regulatory compliance
• High availability
• Independent scaling
• AI-heavy workload

---

1. MICROSERVICE ARCHITECTURE OVERVIEW

Instead of one Laravel monolith, the system is split into independently deployable services.

Each service has:

• Own database schema
• Own Redis instance
• Own API routes
• Independent scaling
• Centralized authentication

1.1 HIGH LEVEL SERVICE MAP

Client (Flutter App)
↓
API Gateway
↓
Service Mesh
↓
Microservices

Core Services:

1. Auth Service
2. User Service

3. Advisor Service
4. Market Sentiment Service
5. Community Service
6. Portfolio Service
7. Finance (UPI) Service
8. Virtual Lab Service
9. AI Service
10. Payment Service
11. Course Service
12. Notification Service
13. Compliance Service
14. Analytics Service

---

2. SERVICE-BY-SERVICE DESIGN

2.1 AUTH SERVICE

Responsibility:
• Login / Register
• JWT generation
• Token validation
• Role-based access control

Database:
auth_db

Tables:
users
roles
refresh_tokens

Endpoints:
POST /login
POST /register
POST /refresh

Scaling:
CPU-light
Horizontally scalable

2.2 USER SERVICE

Responsibility:
• Profile management
• Follow/unfollow
• Preferences

Database:
user_db

Tables:
user_profiles
user_connections

2.3 ADVISOR SERVICE

Responsibility:
• SEBI verification
• Advisor profile
• Advisor statistics

Database:
advisor_db

Tables:
advisor_profiles
verification_documents
advisor_metrics

2.4 MARKET SENTIMENT SERVICE

Responsibility:
• Advisor posts
• Market feed
• Comments
• Likes

Database:
market_db

Tables:
market_posts

market_comments
market_likes

Dependencies:
AI Service
Compliance Service
Notification Service

2.5 COMMUNITY SERVICE

Responsibility:
• General social feed
• Posts
• Comments

Database:
community_db

2.6 PORTFOLIO SERVICE

Responsibility:
• Broker integration
• Holdings storage
• AI trigger

Database:
portfolio_db

Tables:
portfolios
portfolio_assets
portfolio_ai_analysis

Dependencies:
AI Service

2.7 FINANCE SERVICE (UPI)

Responsibility:
• UPI connection

• Transaction fetch
• Budget logic

Database:
finance_db

Tables:
transactions
categories
budgets

Security:
High encryption
Token vault storage

2.8 VIRTUAL LAB SERVICE

Responsibility:
• Virtual trading
• Leaderboard

Database:
lab_db

Tables:
virtual_accounts
virtual_trades

2.9 AI SERVICE

Technology:
Python + FastAPI (recommended)

Responsibilities:
• Portfolio analysis
• Expense optimization
• Sentiment compliance scoring
• Chatbot

Database:
ai_db

Stateless preferred
Cache-heavy

2.10 PAYMENT SERVICE

Responsibility:
• Payment gateway integration
• Webhook handling
• Subscription management

Database:
payment_db

Tables:
transactions
subscriptions
invoices

2.11 COURSE SERVICE

Responsibility:
• Course creation
• Enrollment
• Content delivery

Database:
course_db

2.12 NOTIFICATION SERVICE

Responsibility:
• Push
• Email
• In-app notifications

Uses:
Message Queue

2.13 COMPLIANCE SERVICE

Responsibility:
• AI moderation
• Audit logs
• Regulatory record retention

Database:
compliance_db

2.14 ANALYTICS SERVICE

Responsibility:
• User metrics
• Engagement metrics
• Risk metrics
• Revenue reports

Connected to Data Warehouse.

---

3. COMMUNICATION PATTERN

3.1 API Gateway

Single entry point.

Handles:
• JWT validation
• Rate limiting
• Logging
• Routing

3.2 Synchronous Communication

REST calls between services.

Used when:
Immediate response required.

Example:
Portfolio Service → AI Service (risk scoring)

3.3 Asynchronous Communication

Message Queue (Redis / Kafka)

Used when:
• Notification dispatch
• AI background processing
• Analytics tracking

Example:
Market Post Created
→ Event emitted
→ Notification Service consumes
→ Send push

---

4. DATABASE STRATEGY

Each service owns its database.

NO cross-service joins.

Cross-service data fetched via API.

Example:
Portfolio Service does NOT directly query user_db.

---

5. SERVICE DISCOVERY

Use:

• Kubernetes Service Discovery
or
• Service Mesh (Istio)

---

6. FAILURE ISOLATION

If AI Service crashes:
• Market Service still works
• Portfolio stores data
• AI reprocessing queued

If Payment Service fails:
• Social modules unaffected

This ensures resilience.

---

7. SCALING STRATEGY PER SERVICE

Service	Scaling Driver
Auth	Login spikes
Market	Feed traffic
Portfolio	Sync operations
AI	Heavy compute
Finance	Transaction bursts
Payment	Subscription renewal cycle
Notification	Push spikes

---

8. OBSERVABILITY

Each service includes:

• Structured logging
• Health endpoint
• Metrics endpoint
• Tracing (OpenTelemetry)

---

9. SECURITY ARCHITECTURE

- JWT signed centrally
- Internal service-to-service authentication
- Encrypted secrets
- Role-based permission enforcement
- Audit trail in Compliance Service

Complete Database Schema

Database: PostgreSQL / MySQL
Architecture: Modular with financial-data isolation
Encryption: AES-256 for sensitive financial columns

---

1. USER & AUTHENTICATION TABLES

1.1 users

Field	Type	Description
id	BIGINT (PK)	Unique user ID
uuid	VARCHAR(36)	Public identifier
full_name	VARCHAR(150)	User full name
email	VARCHAR(150)	Unique email
phone	VARCHAR(20)	Mobile number
password	VARCHAR(255)	Hashed password
role_id	BIGINT (FK)	Reference to roles
status	ENUM(active, suspended, pending)	Account state
email_verified_at	TIMESTAMP	Verification timestamp
created_at	TIMESTAMP	Record creation
updated_at	TIMESTAMP	Last update

1.2 roles

Field	Type	Description
id	BIGINT (PK)	Role ID
role_name	VARCHAR(50)	Admin / Advisor / User

1.3 kyc_documents

Field	Type	Description
id	BIGINT (PK)	Document ID
user_id	BIGINT (FK)	Linked user
document_type	ENUM(PAN, Aadhaar, SEBI_CERT)	Document type
document_number	VARCHAR(100)	Encrypted
document_file	TEXT	File path
verification_status	ENUM(pending, approved, rejected)	Status
verified_at	TIMESTAMP	Approval date

1.4 consent_logs

Field	Type	Description
id	BIGINT (PK)	Consent record
user_id	BIGINT (FK)	Linked user
consent_type	VARCHAR(100)	Portfolio / UPI
granted_at	TIMESTAMP	Grant time
revoked_at	TIMESTAMP	Revocation time

---

2. ADVISOR & SENTIMENT MODULE

2.1 advisors

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)

sebi_registration_number	VARCHAR(100)
experience_years	INT
bio	TEXT
verification_status	ENUM(pending, approved, rejected)
created_at	TIMESTAMP

2.2 sentiments

Field	Type
id	BIGINT (PK)
advisor_id	BIGINT (FK)
title	VARCHAR(200)
content	TEXT
market_symbol	VARCHAR(50)
sentiment_type	ENUM(bullish, bearish, neutral)
confidence_score	DECIMAL(5,2)
compliance_status	ENUM(pending, approved, flagged)
created_at	TIMESTAMP

2.3 sentiment_aggregates

Field	Type
id	BIGINT (PK)
market_symbol	VARCHAR(50)
bullish_count	INT
bearish_count	INT

neutral_count	INT
aggregated_score	DECIMAL(5,2)
updated_at	TIMESTAMP

---

3. SOCIAL COMMUNITY TABLES

3.1 social_posts

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
content	TEXT
media_url	TEXT
post_type	ENUM(text, image, video)
visibility	ENUM(public, group)
created_at	TIMESTAMP

3.2 comments

Field	Type
id	BIGINT (PK)
post_id	BIGINT (FK)
user_id	BIGINT (FK)
comment_text	TEXT
created_at	TIMESTAMP

3.3 groups

Field	Type
id	BIGINT (PK)
group_name	VARCHAR(150)
description	TEXT
created_by	BIGINT (FK)

3.4 group_members

Field	Type
id	BIGINT (PK)
group_id	BIGINT (FK)
user_id	BIGINT (FK)
joined_at	TIMESTAMP

3.5 reputation_logs

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
points	INT
activity_type	VARCHAR(100)
created_at	TIMESTAMP

---

4. PORTFOLIO MODULE

4.1 broker_accounts

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
broker_name	VARCHAR(100)
oauth_token	TEXT (Encrypted)
refresh_token	TEXT (Encrypted)
created_at	TIMESTAMP

4.2 portfolios

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
total_value	DECIMAL(15,2)
risk_score	DECIMAL(5,2)
last_synced_at	TIMESTAMP

4.3 holdings

Field	Type
id	BIGINT (PK)
portfolio_id	BIGINT (FK)
symbol	VARCHAR(50)
quantity	DECIMAL(15,4)
average_price	DECIMAL(15,2)
current_price	DECIMAL(15,2)

sector	VARCHAR(100)

4.4 trades_real

Field	Type
id	BIGINT (PK)
portfolio_id	BIGINT (FK)
symbol	VARCHAR(50)
trade_type	ENUM(buy, sell)
quantity	DECIMAL(15,4)
price	DECIMAL(15,2)
trade_date	TIMESTAMP

---

5. VIRTUAL INVESTMENT LAB

5.1 virtual_wallets

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
balance	DECIMAL(15,2)
created_at	TIMESTAMP

5.2 trades_virtual

Field	Type

id	BIGINT (PK)
wallet_id	BIGINT (FK)
symbol	VARCHAR(50)
trade_type	ENUM(buy, sell)
quantity	DECIMAL(15,4)
price	DECIMAL(15,2)
trade_date	TIMESTAMP

5.3 leaderboard

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
roi_percentage	DECIMAL(6,2)
rank_position	INT
period	VARCHAR(50)

---

6. PERSONAL FINANCE & UPI MODULE

6.1 bank_accounts

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
bank_name	VARCHAR(150)
account_masked	VARCHAR(50)

aa_token	TEXT (Encrypted)
created_at	TIMESTAMP

6.2 transactions

Field	Type
id	BIGINT (PK)
bank_account_id	BIGINT (FK)
amount	DECIMAL(15,2)
transaction_type	ENUM(debit, credit)
merchant_name	VARCHAR(150)
category_id	BIGINT (FK)
transaction_date	TIMESTAMP

6.3 expense_categories

Field	Type
id	BIGINT (PK)
category_name	VARCHAR(100)
parent_category	VARCHAR(100)

6.4 budgets

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
category_id	BIGINT (FK)

monthly_limit	DECIMAL(15,2)
month	VARCHAR(20)

6.5 financial_scores

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
score	INT
calculation_date	TIMESTAMP

---

7. AI & COMPLIANCE TABLES

7.1 ai_logs

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
module	VARCHAR(100)
input_data	JSON
output_data	JSON
created_at	TIMESTAMP

7.2 compliance_logs

Field	Type

id	BIGINT (PK)
reference_id	BIGINT
module	VARCHAR(100)
status	ENUM(approved, flagged)
notes	TEXT
created_at	TIMESTAMP

---

8. MONETIZATION TABLES

8.1 courses

Field	Type
id	BIGINT (PK)
advisor_id	BIGINT (FK)
title	VARCHAR(200)
description	TEXT
price	DECIMAL(10,2)
video_url	TEXT

8.2 subscriptions

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
advisor_id	BIGINT (FK)
amount	DECIMAL(10,2)

start_date	TIMESTAMP
end_date	TIMESTAMP
status	ENUM(active, expired)

8.3 payments

Field	Type
id	BIGINT (PK)
user_id	BIGINT (FK)
payment_type	VARCHAR(100)
amount	DECIMAL(10,2)
transaction_id	VARCHAR(200)
status	ENUM(success, failed)
created_at	TIMESTAMP

---

9. INDEXING STRATEGY

Table	Indexed Fields
users	email, phone
sentiments	market_symbol, sentiment_type
holdings	portfolio_id, symbol
transactions	transaction_date, category_id
trades_virtual	wallet_id
social_posts	user_id

---

10. Data Security Strategy

- Encrypted columns:
    - oauth_token
    - aa_token
    - document_number
- Role-based access constraints
- Separate logical schema for financial tables
- AI logs anonymized where required

SEQUENCE DIAGRAMS

1.1 Market Sentiment Post – Advisor Flow

Actors:
Advisor → Flutter App → API → AI Compliance → DB → Notification Service

Sequence:

Advisor
→ Submit Post

Flutter App
→ POST /market/posts

API Server
→ Validate JWT
→ Validate Advisor Status
→ Validate Disclaimer

API
→ Send Content to AI Compliance Service

AI Service
→ Analyze manipulation risk
→ Return Risk Score + Flag Status

IF Flagged:
API → Store status = under_review
API → Notify Admin
ELSE:
API → Insert into market_posts
API → Trigger Notification Service

Notification Service
→ Create notification records
→ Push to followers

Response returned to Advisor.

1.2 Portfolio AI Analysis Flow

User → Flutter → API → Portfolio Service → AI Engine → DB

Sequence:

User
→ Connect Broker

Flutter
→ OAuth Flow

Broker API
→ Return Access Token

API
→ Encrypt token
→ Fetch holdings

API
→ Save holdings

API
→ Trigger AI Analysis Job (Queue)

AI Engine
→ Calculate:

- Allocation %
- Volatility score
- Concentration ratio
- Beta exposure

AI Engine
→ Generate:

- Risk Score
- Diversification Score
- Rebalance Suggestion

AI Engine
→ Save in portfolio_ai_analysis

API
→ Notify User (High Risk if applicable)

1.3 Virtual Trade Execution Flow

User → Flutter → API → Price Service → DB → Leaderboard

User
→ Place Buy Order

API
→ Validate virtual balance
→ Fetch live price

Price API
→ Return current price

API
→ Calculate total cost
→ Deduct balance
→ Insert virtual_trade record
→ Update virtual_account balance

API
→ Recalculate leaderboard

1.4 AI Chatbot Flow

User → API → Context Builder → AI Service → DB → Response

User
→ Ask Question

API
→ Identify Query Type

If Portfolio Query:
Fetch portfolio data

If Expense Query:
Fetch spending summary

API
→ Construct Structured Prompt

AI Service
→ Generate response

API
→ Save ai_chats record
→ Return response

---

2. TECHNICAL ARCHITECTURE DIAGRAM (LAYERED DESIGN)

2.1 High-Level Architecture

Client Layer
→ Flutter App (Android + iOS)

API Layer
→ Laravel REST API
→ JWT Middleware
→ Rate Limiter

Business Logic Layer
→ Market Module
→ Portfolio Module
→ Finance Module
→ AI Module
→ Course Module

Service Layer
→ AI Microservice (Python - optional)
→ Broker Integration Service
→ Notification Service

Data Layer
→ MySQL
→ Redis Cache
→ Queue System

External Integrations
→ Broker APIs
→ UPI APIs
→ Payment Gateway
→ LLM Provider
→ Market Data Provider

2.2 Modular Breakdown

Monolith (Phase 1)

Future Microservices (Phase 2+):

Auth Service
Market Service
Portfolio Service
Finance Service
AI Service
Notification Service

Each with:
Own DB schema
Own Redis cache
Independent scaling

---

3. AI ALGORITHM LOGIC SPECIFICATION

This section defines internal logic, not just LLM usage.

3.1 Portfolio Risk Score Algorithm

Inputs:

- Equity %
- Small cap %
- Crypto %
- Sector concentration
- Volatility index
- Number of assets

Risk Score Formula (Example Weighted Model):

Risk Score =
(Equity% × 0.25) +
(SmallCap% × 0.20) +
(Crypto% × 0.20) +
(Concentration Ratio × 0.15) +
(Volatility Score × 0.20)

Output:
0–3 Low
4–6 Moderate
7–10 High

3.2 Diversification Score

Diversification =
1 − (Top 3 Assets Weight / Total Portfolio)

Higher score = better diversification.

3.3 Expense Optimization AI

Inputs:

- Monthly income
- Category spending
- Historical pattern

Logic:
Detect:

- Sudden spike
- Recurring subscription
- High discretionary spending

Output:
Suggestions:

- Reduce eating out by 15%
- Consolidate subscriptions
- Increase SIP by surplus

3.4 Sentiment Compliance AI

Rule-Based + LLM Hybrid

Rules:
Block phrases like:

- “Guaranteed returns”
- “100% safe”
- “No risk”
- “Buy now before it explodes”

LLM:
Evaluate tone + intent.

Output:
Risk Score:
Low
Moderate
High (flag to admin)

3.5 Rebalance Suggestion Logic

Target Allocation (Example Model):
Equity 60%
Debt 30%
Gold 10%

If actual allocation deviates >10%, suggest rebalance.

---

4. COMPLIANCE & RISK MAPPING MATRIX

Risk Area	Risk Type	Mitigation
Market Manipulation	Pump & dump	AI compliance scan
Unverified Advisor	Fraud risk	Manual approval
Misleading Claims	Legal liability	Mandatory disclaimers
Data Breach	Financial data risk	Encryption
AI Hallucination	Wrong advice	Confidence scoring
UPI Data Leak	Sensitive financial info	Token encryption
Broker Token Theft	Account compromise	AES encryption

Audit Requirements:

- Log all advisor posts
- Log edits
- Store moderation history
- Store AI compliance scores

---

5. DEVOPS & DEPLOYMENT FLOW

5.1 CI/CD Pipeline

Developer
→ Push to GitHub

GitHub Actions:
→ Run tests
→ Build Docker image
→ Deploy to staging

Manual Approval
→ Deploy to production

5.2 Environment Setup

Development
Staging
Production

Each has:
Separate DB
Separate Redis
Separate environment configs

5.3 Deployment Architecture

Load Balancer
→ NGINX
→ Laravel App Instances (Auto-scaled)
→ Redis
→ MySQL Primary + Replica

5.4 Backup Strategy

Daily DB backups
Weekly full snapshot
Encrypted S3 storage

5.5 Monitoring

Tools:

- Server monitoring
- Error logging
- API latency tracking

- Queue monitoring
- AI service uptime tracking

COMPLETE SYSTEM FLOW SUMMARY

User
→ Flutter
→ API Gateway
→ Business Logic
→ AI + Compliance
→ DB
→ Notification Engine

Security enforced at every layer.

API Endpoint Documentation

---

1. API ARCHITECTURE OVERVIEW

Base URL

https://api.yourapp.com/v1/

Authentication

- JWT-based Authentication
- OAuth2 for external broker integrations
- API Rate Limiting (especially for market sentiment posts)

Headers

Header	Required	Description
Authorization	Yes	Bearer {JWT_TOKEN}
Content-Type	Yes	application/json
X-Device-ID	Yes	Unique device identifier
X-App-Version	Yes	For version control

---

2. AUTHENTICATION & USER MANAGEMENT APIs

2.1 Registration

POST `/auth/register`

Field	Type	Required
name	string	Yes
email	string	Yes
phone	string	Yes

password	string	Yes
role	enum (user/advisor/admin)	Yes
sebi_registration_no	string	Required if advisor

Response:

```
{
  "token": "",
  "user": {}
}
```

2.2 Login

POST `/auth/login`

Field	Type
email/phone	string
password	string

2.3 Logout

POST `/auth/logout`

2.4 Get Profile

GET `/user/profile`

2.5 Update Profile

PUT `/user/profile`

---

3. SEBI ADVISOR VERIFICATION APIs

3.1 Submit Verification

POST `/advisor/verify`

Field	Type
sebi_registration_no	string
certificate_upload	file
experience_years	integer

3.2 Get Advisor Status

GET `/advisor/status`

---

4. MARKET SENTIMENT SOCIAL FEED APIs

4.1 Create Market Post (Advisors Only)

POST `/market/posts`

Field	Type
title	string
content	text
asset_type	enum (equity, crypto, mf, commodity)
sentiment	enum (bullish, bearish, neutral)
risk_level	enum (low, medium, high)
disclaimer	text

4.2 Fetch Market Feed

GET `/market/posts`

Query Params:

- asset_type
- sentiment
- advisor_id
- page

4.3 Comment on Market Post

POST `/market/posts/{id}/comment`

4.4 Like Market Post

POST `/market/posts/{id}/like`

4.5 Report Post

POST `/market/posts/{id}/report`

---

5. GENERAL SOCIAL COMMUNITY APIs

5.1 Create Social Post

POST `/community/posts`

Field	Type
content	text
media	file

category	enum (general, learning, motivation, news)

5.2 Fetch Community Feed

GET `/community/posts`

5.3 Follow User

POST `/community/follow/{user_id}`

5.4 Unfollow User

DELETE `/community/unfollow/{user_id}`

5.5 Save Post

POST `/community/posts/{id}/save`

---

6. PORTFOLIO CONNECT & ANALYTICS APIs

6.1 Connect Broker Account

POST `/portfolio/connect`

Field	Type
broker_name	string
api_token	string

6.2 Fetch Portfolio

GET `/portfolio`

6.3 Get AI Portfolio Insights

GET `/portfolio/ai-insights`

Response:


```
{
  "risk_score": 7.5,
  "diversification_score": 60,
  "suggested_rebalance": [],
  "sector_exposure": []
}
```

6.4 Manual Portfolio Entry

POST `/portfolio/manual-add`

6.5 Portfolio Performance Chart

GET `/portfolio/performance`

---

7. VIRTUAL INVESTMENT LAB APIs

7.1 Create Virtual Account

POST `/lab/create`

7.2 Fetch Virtual Balance

GET `/lab/balance`

7.3 Place Virtual Trade

POST `/lab/trade`

Field	Type
asset_symbol	string
quantity	integer
trade_type	buy/sell

7.4 Fetch Virtual Portfolio

GET `/lab/portfolio`

7.5 Leaderboard

GET `/lab/leaderboard`

---

8. AI CHATBOT APIs

8.1 Ask Financial Question

POST `/ai/chat`

Field	Type
question	text
context	optional

Response:

{
  "answer": "",
  "confidence_score": 0.92
}

8.2 AI Risk Assessment

POST `/ai/risk-profile`

8.3 AI Expense Insights

GET `/ai/expense-analysis`

---

9. PERSONAL FINANCE & UPI TRACKING APIs

9.1 Connect UPI

POST `/finance/connect-upi`

9.2 Fetch Transactions

GET `/finance/transactions`

9.3 Categorize Transaction

POST `/finance/categorize`

9.4 Budget Setup

POST `/finance/budget`

9.5 Monthly Report

GET `/finance/monthly-report`

---

10. COURSE & SUBSCRIPTION MODULE APIs

10.1 Create Course (Advisor Only)

POST `/courses/create`

10.2 List Courses

GET `/courses`

10.3 Purchase Course

POST `/courses/{id}/purchase`

10.4 Subscribe to Advisor

POST `/advisor/{id}/subscribe`

---

11. NOTIFICATIONS APIs

11.1 Fetch Notifications

GET `/notifications`

11.2 Mark as Read

POST `/notifications/read`

---

12. ADMIN PANEL APIs

12.1 Approve Advisor

POST `/admin/advisor/{id}/approve`

12.2 Moderate Post

POST `/admin/posts/{id}/moderate`

12.3 View Reports

GET `/admin/reports`

---

13. SECURITY & COMPLIANCE ENDPOINTS

13.1 Content AI Moderation

POST `/moderation/analyze`

13.2 Audit Logs

GET `/admin/audit-logs`

---

14. REAL-TIME FEATURES

Use WebSockets for:

- Live market sentiment feed
- Chat
- Notification push
- Leaderboard updates

Endpoint:

wss://api.yourapp.com/ws

---

15. RATE LIMITING STRATEGY

Endpoint Type	Limit
AI Chat	50/day free
Market Posts	5/hour
Comments	30/hour
Portfolio Sync	10/day

---

16. ERROR RESPONSE FORMAT

```
{
  "status": false,
  "message": "Validation error",
  "errors": {
    "email": ["Email already exists"]
  }
}
```

AI & ANALYTICS

AI Agent Architecture

1.1 Architectural Philosophy

The AI layer is designed as:

• Modular
• Multi-agent
• Context-aware
• Compliance-controlled
• Event-driven
• Explainable
• Horizontally scalable

It must NEVER act as a direct investment advisor unless legally permitted.
It provides informational, analytical, and risk-based insights.

1.2 High-Level AI Architecture

Client (Flutter)
→ AI Gateway
→ AI Orchestrator
→ Task Classifier
→ Specialized Agent
→ Data Retrieval Layer (RAG)
→ Response Generator
→ Guardrail Engine
→ Structured Output

1.3 Core Components

1. AI Gateway

Responsibilities:
• Authentication
• Rate limiting
• Context injection
• Logging

Example:
User asks: “Is my portfolio risky?”
Gateway injects:

- User risk profile
- Portfolio snapshot
- Recent AI insights

---

2. AI Orchestrator

Central controller that:

• Classifies query type
• Routes to correct agent
• Manages fallback logic
• Aggregates multi-agent responses

Example:
User asks:
“How much am I overspending and should I rebalance?”

Orchestrator splits into:

- Expense Agent
- Portfolio Agent

Then merges responses.

---

3. Task Classification Engine

Categories:

Category	Agent Triggered
Portfolio analysis	Portfolio Agent

Expense advice	Expense Agent
Market explanation	Financial Q&A Agent
Sentiment scoring	Sentiment Agent
Content moderation	Compliance Agent
Fraud detection	Risk Agent

1.4 Specialized AI Agents

A. Portfolio Insight Agent

Inputs:
- Portfolio assets
- Allocation data
- Market volatility data
- User risk profile

Outputs:
- Risk score
- Diversification warnings
- Rebalancing suggestions
- Exposure alerts

Example Output:

"You are 48% exposed to financial services. This exceeds recommended 30% for moderate risk investors."

Model Types:
- Risk scoring ML model
- Volatility clustering model
- Sector exposure threshold model

B. Expense Intelligence Agent

Inputs:
- UPI transactions

- Budget settings
- Historical spending patterns

Outputs:
- Category analysis
- Budget breach alerts
- Savings trend forecast

Models:
- Transaction categorization (NLP)
- Time-series forecasting
- Anomaly detection

Example:

“Dining expenses increased 34% compared to last month.”

C. Sentiment Analysis Agent

Used in:

- Market posts
- Trend detection
- Manipulation risk

Outputs:
- Sentiment classification
- Confidence score
- Volatility correlation score

Models:
- Financial sentiment NLP
- Abnormal pattern detection
- Advisor historical accuracy ML

D. Compliance Moderation Agent

Used before publishing advisor posts.

Checks for:
- Guaranteed return claims
- Insider trading hints

• Pump-and-dump behavior
• Unrealistic targets

Output:
• Approved
• Flagged
• Rejected
• Escalate to Admin

E. Financial Q&A Agent

LLM-based agent with:

• RAG (Retrieval Augmented Generation)
• SEBI guidelines injection
• Portfolio-aware context

Restrictions:
• Cannot give “Buy X stock now”
• Must avoid price prediction certainty

F. Fraud & Manipulation Detection Agent

Monitors:

• Sudden coordinated bullish posts
• Engagement farming
• Fake account clustering
• Payment abuse

Uses:
• Graph-based anomaly detection
• Behavioral clustering
• Velocity checks

1.5 Guardrail Engine

Final layer before response delivery.

Checks:

- Regulatory language
- No financial guarantees
- No direct stock tip
- No portfolio trade execution advice
- No hallucinated numbers

1.6 Explainability Layer

Every AI response stores:

- Input snapshot
- Model version
- Confidence score
- Decision explanation
- Risk category

Stored in:
ai_decision_logs table

AI Dependency Grid

2.1 Feature-to-Agent Mapping

Feature	AI Agent Used	Dependency Level	Output Type
Portfolio Risk Score	Portfolio Agent	High	Score + Insight
Asset Allocation Warning	Portfolio Agent	High	Risk Alert
Rebalancing Suggestion	Portfolio Agent	High	Text + Metrics
Expense Categorization	Expense Agent	High	Category Label
Budget Breach Alert	Expense Agent	Medium	Notification
Market Post Sentiment	Sentiment Agent	High	Bullish/Bearish
Advisor Accuracy Tracking	Sentiment Agent	High	% Score
Manipulation Monitoring	Fraud Agent	Critical	Admin Alert
Post Moderation	Compliance Agent	Critical	Approve/Reject
Chatbot Query	Q&A Agent	High	Conversational Response
Trend Detection	Sentiment Agent	Medium	Trending List
Subscription Fraud	Fraud Agent	High	Risk Score
Engagement Bot Detection	Fraud Agent	High	Account Flag

2.2 Cross-Agent Dependencies

Portfolio Agent depends on:
- Market data
- Risk profile
- Snapshot engine

Expense Agent depends on:
- Transaction feed
- Budget data

Compliance Agent depends on:
- NLP model
- SEBI rule dataset
- Market history

Fraud Agent depends on:
- User graph
- Engagement logs
- Payment logs

Compliance AI Mapping

4.1 Regulatory Safeguards

The system ensures:

- Advisors must be verified
- No guaranteed returns
- Historical transparency
- Manipulation prevention
- Audit-ready logs

4.2 Compliance AI Workflow

Advisor Post
→ NLP Scan
→ Rule Engine Check
→ Historical Pattern Analysis
→ Volatility Correlation
→ Approve/Flag

4.3 Rule Engine Conditions

Flags if post contains:

- “Guaranteed returns”
- “100% profit”
- “Target ₹500 in 2 days”
- Insider-like hints
- Coordinated bullish surge

4.4 Manipulation Risk Scoring

Risk Score Formula:

Score =
Engagement Velocity +
New Account Participation +
Sentiment Spike +
Market Price Movement Correlation

If threshold exceeded:
→ Freeze post visibility
→ Alert admin
→ Log event

4.5 Advisor Transparency Mechanism

Publicly visible:

• Accuracy %
• Historical sentiment chart
• Number of predictions
• Risk level

4.6 AI Audit Log Structure

ai_decision_logs

- id
- user_id
- agent_type
- input_data_hash
- output_summary
- risk_flag
- confidence_score
- model_version
- created_at

Required for regulatory inspection.